CHAPTER III: METHODOLOGY

3.1 Introduction

This chapter presents the methodology used in the design and development of the proposed Police Station Management System (PSMS), also known as the Criminal Case Management System (CMS). The chapter explains the techniques, architectural design, development tools, security mechanisms, functional modules, database structure, and implementation approach used during the development process.

The proposed system is designed to modernize police station operations by replacing manual paper-based procedures with a centralized digital platform capable of managing criminal cases, officer records, evidence tracking, investigation workflows, and administrative operations. The methodology adopted ensures that the system remains secure, scalable, efficient, and suitable for modern law enforcement environments.

The system integrates advanced technologies such as blockchain-based integrity verification, role-based access control, audit logging, and secure authentication mechanisms to ensure confidentiality, accountability, and transparency in police operations. Similar criminal records and police management methodologies emphasize structured workflows, centralized information systems, and secure database management for law enforcement institutions. (Qeios)

3.2 System Overview

The Police Station Management System is a secure web-based criminal case management platform developed to support law enforcement agencies in managing police operations digitally. The system is designed using a hierarchical administrative structure that ensures proper supervision, controlled data visibility, and secure access to criminal records.

The system operates using a five-tier administrative hierarchy consisting of:

1. State Administration
2. Regional Administration
3. City Administration
4. District Administration
5. Neighborhood / Police Station Level

This hierarchical model ensures that users can only access information related to their jurisdiction and subordinate administrative levels. Peer-level access restrictions prevent unauthorized viewing of records belonging to other regions or districts.

The system automates the complete criminal case workflow beginning from incident reporting and Occurrence Book (OB) generation to investigation, prosecution referral, and case closure. The platform also supports evidence tracking, officer deployment management, chain of custody recording, blockchain integrity verification, and audit logging.

The system is accessible through modern web browsers and can operate on desktop computers, laptops, tablets, and smartphones. The centralized structure improves operational efficiency, reduces paperwork, strengthens data security, and improves coordination between police departments and specialized investigation units.

The proposed system also incorporates blockchain-based cryptographic verification using SHA-256 hashing algorithms to prevent unauthorized modification of critical case records. Similar approaches have been recommended in modern criminal database systems to improve integrity protection and accountability. (KIU Institutional Repository)

3.3 System Features

The proposed Police Station Management System contains multiple integrated modules that collectively support law enforcement activities and administrative management.

3.3.1 User Authentication and Access Control

The system allows secure user registration, login, and logout functionalities. Access is controlled through role-based authorization mechanisms ensuring that users only access permitted modules according to their roles and jurisdictions.

3.3.2 Administrative Hierarchy Management

The system manages five administrative levels including State, Region, City, District, and Station. The hierarchy ensures proper command structures and restricted visibility across organizational units.

3.3.3 Occurrence Book (OB) Management

The platform automatically generates unique OB numbers for every reported incident. OB records contain complaint details, timestamps, assigned officers, and case references.

3.3.4 Criminal Case Management

The system manages the entire lifecycle of criminal cases including:

- Draft Stage
- Commander Review
- Investigation
- CID Referral
- Prosecutor Referral
- Court Review
- Case Closure

3.3.5 Complainant Management

The module stores complainant information including personal details, contact information, identification records, and submitted complaints.

3.3.6 Victim Management

The system records victim profiles, statements, medical reports, and related case evidence.

3.3.7 Suspect Management

The suspect module stores criminal suspect records including personal information, charges, photographs, arrest details, and criminal history.

3.3.8 Witness Management

Witness statements and contact details are securely stored to support investigations and court proceedings.

3.3.9 Evidence Management System

The evidence module manages physical and digital evidence linked to criminal investigations. Evidence categories include documents, photographs, videos, fingerprints, and seized items.

3.3.10 Chain of Custody Tracking

The chain of custody module tracks all evidence transfers between officers and departments. Each movement is recorded with timestamps, officer identities, and transfer descriptions.

3.3.11 Officer Registry Management

The system maintains a centralized registry of police officers including:

- Force Number
- Rank
- Deployment Location
- Transfer History

- Service Status

3.3.12 Officer Deployment and Transfer Management

The platform supports reassignment and transfer of officers across administrative levels while maintaining historical deployment records.

3.3.13 CID Investigation Module

Complex criminal cases can be referred to Criminal Investigation Department (CID) officers for advanced investigations and specialized handling.

3.3.14 Prosecutor and Court Module

Approved criminal cases can be forwarded to prosecutors and court officers for legal review and prosecution management.

3.3.15 Jail and Corrections Module

The system tracks arrested suspects, detention records, custody periods, and correctional status updates.

3.3.16 Blockchain Integrity Verification

Critical case events generate SHA-256 cryptographic hashes stored in blockchain records to prevent tampering and unauthorized modifications.

3.3.17 Audit Logging System

Every system activity including login attempts, updates, evidence uploads, transfers, and deletions is recorded in the audit trail.

3.3.18 Dashboard and Analytics

The dashboard provides crime statistics, investigation summaries, officer counts, evidence reports, and operational analytics in real time.

3.3.19 Report Generation

The system generates printable and exportable reports including:

- Crime Reports
- Investigation Reports
- Officer Reports
- Evidence Reports
- Court Referral Reports

3.3.20 Notification and Alert System

Automated notifications alert officers and administrators regarding case updates, transfer approvals, pending investigations, and evidence submissions.

3.4 System Architecture

The Police Station Management System follows a Three-Tier Architecture Model consisting of:

1. Presentation Layer
2. Application Layer
3. Data Layer

This architecture improves maintainability, scalability, and security.

Presentation Layer

The presentation layer provides graphical user interfaces for police officers, investigators, prosecutors, administrators, and correctional officers.

The frontend is developed using:

- Next.js
- React.js
- Ant Design (AntD)

The interface is responsive and accessible through modern web browsers on multiple devices.

Application Layer

The application layer contains the business logic of the system. It processes requests, validates user permissions, manages workflows, and handles system operations.

Technologies used include:

- Node.js
- Express.js
- RESTful APIs

Core functionalities implemented in this layer include:

- JWT Authentication
- Role-Based Access Control
- Blockchain Hash Generation
- Audit Logging
- Case Processing

Data Layer

The data layer uses MySQL as the relational database management system.

The database stores:

- User Accounts
- Criminal Cases
- Evidence Records
- Officer Information
- Blockchain Hashes
- Audit Logs
- Transfer Histories

Indexes and foreign key constraints are implemented to improve query performance and maintain referential integrity.

Modern police information management systems commonly apply centralized database architectures and layered designs to improve scalability and operational efficiency. (Office of Justice Programs

3.5 System Requirements

3.5.1 Hardware Requirements

No	Component	Specification
1	Processor	Intel Core i5 or AMD Ryzen 5
2	RAM	Minimum 8 GB DDR4
3	Storage	256 GB SSD
4	Display	14-inch Full HD Monitor
5	Internet	Stable Broadband Connection
6	Input Devices	Keyboard and Mouse
7	Backup Device	External Storage Device

3.5.2 Software Requirements

No	Software	Purpose
1	Ubuntu Linux	Server Operating System
2	Node.js	Backend Runtime
3	Express.js	API Framework
4	Next.js	Frontend Framework
5	React.js	User Interface
6	MySQL	Relational Database
7	JWT	Authentication
8	bcrypt	Password Encryption
9	Git	Version Control
10	Visual Studio Code	Development Environment
11	Postman	API Testing
12	Google Chrome	Browser Testing

3.6 Functional and Non-Functional Requirements

3.6.1 Functional Requirements

R1

The system shall allow secure user authentication using usernames and passwords.

R2

The system shall generate unique OB numbers for all incidents.

R3

The system shall support criminal case registration and management.

R4

The system shall allow evidence upload and tracking.

R5

The system shall maintain chain of custody records.

R6

The system shall support officer deployment and transfers.

R7

The system shall allow CID officers to manage referred cases.

R8

The system shall support prosecutor and court review processes.

R9

The system shall generate operational reports.

R10

The system shall log all user activities for auditing purposes.

R11

The system shall generate blockchain hashes for critical events.

R12

The system shall support hierarchical data visibility restrictions.

3.6.2 Non-Functional Requirements

NFR1 — Security

The system must secure sensitive information using JWT authentication and bcrypt password hashing.

NFR2 — Performance

The system should process requests efficiently with minimal response time.

NFR3 — Scalability

The system should support increasing numbers of users and criminal records.

NFR4 — Reliability

The platform must maintain data consistency and availability.

NFR5 — Availability

The system should be accessible continuously through web browsers.

NFR6 — Maintainability

The application architecture should support future modifications and updates.

NFR7 — Integrity

Blockchain verification mechanisms must protect critical records from tampering.

NFR8 — Usability

The interface should remain simple and understandable for police personnel.

3.7 Feasibility Study

Technical Feasibility

The project is technically feasible because it uses modern web technologies including Node.js, Express.js, React.js, Next.js, and MySQL. These technologies are widely supported and suitable for enterprise-level web applications. Similar police systems and criminal databases have successfully adopted centralized digital platforms and layered architectures. (KIU Institutional Repository)

Economic Feasibility

The system minimizes operational costs by reducing paperwork, manual record keeping, and administrative delays. Since most technologies used are open-source, licensing costs are minimal.

Operational Feasibility

The system is operationally feasible because police officers and administrators can easily use the web interface with minimal technical training.

Schedule Feasibility

The project can be completed within the proposed academic timeline using incremental development and modular implementation approaches. Incremental methodologies are commonly used in crime management system development projects. (SuPlus)

3.8 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the proposed system.

3.8.1 Use Case Diagram

The Use Case Diagram illustrates interactions between:

- System Administrator
- Police Officer
- CID Investigator
- Prosecutor
- Jail Officer

and the modules they can access.

3.8.2 Activity Diagram

The Activity Diagram illustrates the sequence of operations involved in criminal case registration, investigation, evidence handling, and prosecution.

3.8.3 Sequence Diagram

The Sequence Diagram shows communication between users, APIs, and the database during system operations.

3.8.4 Entity Relationship Diagram (ERD)

The ERD defines relationships between system entities including users, officers, cases, evidence, suspects, witnesses, and blockchain records.

3.8.5 Role-Based Access Control Diagram

This diagram illustrates role permissions and hierarchical access restrictions across administrative levels.

Research on crime management systems commonly recommends UML modeling techniques including ERDs, use case diagrams, and activity diagrams for structured software design and criminal workflow analysis. (IIETA)

3.9 Database Design

The proposed system uses MySQL as the relational database management system. The database is normalized to reduce redundancy and improve consistency.

The database contains multiple interconnected tables including:

Table Name	Purpose
Users	Stores system users and roles
Officers	Stores police officer records
Cases	Stores criminal case information
OB_Records	Stores occurrence book entries
Evidence	Stores evidence details
Chain_of_Custody	Tracks evidence transfers
Witnesses	Stores witness records
Suspects	Stores suspect records
Victims	Stores victim information
Audit_Logs	Stores user activity logs
Blockchain_Records	Stores SHA-256 hashes
Transfers	Stores officer transfer history

Table Name	Purpose
Notifications	Stores system alerts
Prosecutor_Reviews	Stores legal reviews
Jail_Records	Stores detention information

Primary keys and foreign key relationships are implemented to maintain referential integrity. Indexes are applied to frequently queried columns to improve performance.

Centralized police databases and criminal information systems typically rely on relational database designs for maintaining secure and structured criminal records. (KIU Institutional Repository)

3.10 Chapter Summary

This chapter presented the methodology used in the design and development of the Police Station Management System. The chapter explained the system overview, architecture, major features, requirements, feasibility study, UML diagrams, and database design.

The proposed system provides a secure and scalable platform capable of improving criminal case management, evidence tracking, officer coordination, administrative monitoring, and organizational accountability within police institutions. The integration of blockchain verification, hierarchical access control, and audit logging ensures high levels of integrity, transparency, and operational security.